

# Empirical Proof Record: VALIDATED

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## Empirical Proof Record — VALIDATED

**Proof ID:** 3396ea02dcc1086d975f2d1e6837b74...

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### 1. Claim

Across 100 random  $(k, n)$  binomial pairs (seed=20260518, max\_n=1000), scipy's "binomtest.proportion\_ci(method='wilson')" and statsmodels's "proportion\_confint(method='wilson')" produce 95% CI bounds that agree to  $\leq 1e-09$  on both lower and upper endpoints. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer.)

- **Operationalisation:** Generate  $N$  pairs  $(k, n)$  with  $k \sim \text{Uniform}(0, n)$ ,  $n \sim \text{Uniform}(2, \text{max\_n})$ . For each pair, compute the Wilson CI at the configured confidence under both backends. Compute  $\text{max\_abs\_diff} = \max$  over pairs of  $\max(|\text{low\_scipy} - \text{low\_sm}|, |\text{high\_scipy} - \text{high\_sm}|)$ . VALIDATED iff  $\text{max\_abs\_diff} \leq \text{tolerance}$ .
- **Threshold:**  $\text{max\_absolute\_ci\_difference} \leq 1e-09$  proportion
- **H0:** scipy and statsmodels disagree on the Wilson CI by more than the floating-point tolerance — at least one implementation has a defect.
- **H1:** Both libraries evaluate the same closed-form Wilson CI and produce identical bounds to floating-point tolerance.

### 2. Verdict

**VALIDATED** — observed 3.3306690738754696e-16

100 pairs checked;  $\max |\text{low\_scipy} - \text{low\_sm}| = 1.110e-16$ ;  $\max |\text{high\_scipy} - \text{high\_sm}| = 3.331e-16$ ;  $\max$  overall = 3.331e-16 ( $\leq 1e-09$  tol)

### 3. Evidence

| Pillar                         | Statistic                  | Value                  | 95% C |
|--------------------------------|----------------------------|------------------------|-------|
| wilson_ci_scipy_vs_statsmodels | max_absolute_ci_difference | 3.3306690738754696e-16 | -     |

### 4. Pre-registration

- Registered at: 2026-05-31T18:33:59.512971+00:00
- Config hash: ab8fb015887c4a2ac85c8b2ffe1e676a...
- Data hash: ab8fb015887c4a2ac85c8b2ffe1e676a...

### 5. Signature

6336af58b41529201f79229d28bf0759e08b81e2702510ec7467871dd93c30c6